

Hammad Ahmad

Data Scientist — Predictive Modeling · Statistical ML · Python

+44 7555 856 155 | hammadahmad.ml@gmail.com | Bradford, UK

[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [ORCID](#)

PROFILE

Data scientist with an MSc (Merit) in Applied AI and a first-author peer-reviewed publication benchmarking machine-learning methods. I build and evaluate predictive models on real-world clinical and text data — ensemble methods, stratified cross-validation, and SHAP-based interpretability — and take them through to REST deployment, working fluently across the Python data stack and modern NLP/LLM tooling.

TECHNICAL SKILLS

Machine Learning & Statistics : Predictive modeling, ensemble methods (Random Forest, Gradient Boosting), classification, stratified cross-validation, SHAP interpretability, scikit-learn, PyTorch, TensorFlow, MLflow

Data Analysis : pandas, NumPy, SQL, feature engineering, data preprocessing, model evaluation

NLP & LLMs : LLMs, Retrieval-Augmented Generation (RAG), Sentence Transformers, semantic search, RAGAS evaluation, Neo4j knowledge graphs

Programming & Tools : Python, C++, FastAPI, REST APIs, PostgreSQL, Docker, Git, Linux

EXPERIENCE

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Oct 2025 – Mar 2026

AI / Machine Learning Engineer

Leeds, UK (Remote)

- Contributed to an AI voice-agent system built with Retell AI and FastAPI, developing backend dialogue-flow logic and workflow automation.
- Built asynchronous backend services and lead-tracking automation for an AI-driven outbound communication tool.
- Integrated LLM-based dialogue systems with production backend infrastructure.

University of Bradford

Jan 2025 – Sep 2025

Research Assistant — Machine Learning & LLMs

Bradford, UK

- Built FinLaw-UK, a RAG system pairing an open-source LLM (Mistral 7B) with a Neo4j knowledge graph for reliable UK financial-regulation question answering.
- Improved retrieval quality by fine-tuning Sentence Transformer embeddings and adding cross-encoder re-ranking.
- Benchmarked answer faithfulness and relevance using the automated RAGAS evaluation framework.

COMSATS University Islamabad

Jul 2023 – Jul 2024

Research Assistant — Data Science

Islamabad, Pakistan

- Developed ensemble predictive models (Random Forest, Gradient Boosting) on high-dimensional clinical datasets, reaching ~93% classification accuracy under stratified cross-validation.
- Applied SHAP-based interpretability to surface the most influential predictors and improve model transparency.
- Deployed trained models behind REST APIs for downstream consumption.

PROJECTS

Jobzyl — Multi-Platform Job Aggregator

jobzyl.com

Full-Stack Architecture, Data Pipelines, Responsive UI

- Built a full-stack platform that ingests real-time job listings from multiple sources (LinkedIn, Indeed, Google Jobs) into a single searchable portal.
- Designed the data-parsing pipelines and responsive UI; live and actively maintained.

EDUCATION

University of Bradford

Sep 2024 – Sep 2025

MSc, Applied Artificial Intelligence and Data Analytics — Merit

Bradford, UK

- **Dissertation**: FinLaw-UK — A Graph-Augmented Retrieval Chatbot for Reliable UK Financial Regulation.

COMSATS University Islamabad

Sep 2020 – Jul 2024

BS, Bioinformatics

Islamabad, Pakistan

PUBLICATIONS

Ahmad, H., Khan, M.U., & Azam, M. (2024). Comparative Analysis of Machine Learning Methods for Enhancing Sleep Efficiency and Prediction. *ICSMAI 2024* (Springer). DOI: 10.1007/978-3-031-66854-8_1